Predicting Customer Churn in the Telecommunication Industries Using Machine Learning Techniques

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DECLARATION AND APPROVAL

DECLARATION

I declare that this work has not been previously submitted and approved for the award of a degree by this or any other University. To the best of my knowledge and belief, the proposal contains no material previously published or written by another person except where due reference is made in the proposal itself.

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APPROVAL

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ABSTRACT

Customer churn prediction in the telecommunications sector is a persistent challenge that significantly affects profitability, competitiveness, and long-term sustainability. The study begins by outlining the background and research problem, noting the increasing competition in the sector and the limitations of traditional churn management approaches. It highlights the objectives of identifying key churn predictors, developing a machine learning–based prediction model, evaluating multiple algorithms, and designing a deployable dashboard to support decision-making. A review of relevant literature synthesizes theoretical frameworks such as Customer Relationship Management (CRM) and Technology Adoption Theory, along with recent empirical studies on churn prediction. Prior work demonstrates the potential of machine learning and deep learning models but also reveals gaps related to interpretability, profit-sensitive evaluation, and limited focus on other telecommunications markets. The methodology follows the CRISP-DM framework and propose to utilize secondary data from the Iranian Churn (UCI) and Duke Cell2Cell (Teradata) datasets. Data preprocessing steps include cleaning, feature engineering, class imbalance handling, and encoding. Several models, including Logistic Regression, Decision Trees, Random Forest, and LightGBM, will be compared, with hyperparameter tuning and interpretability techniques such as SHAP applied to enhance performance and stakeholder trust. Expected outcomes include the development of a robust churn prediction system with high predictive accuracy, interpretability, and scalability. From an operational perspective, the system will enable proactive customer retention strategies, optimized resource allocation, and continuous monitoring. This research contributes a deployable and replicable framework that addresses the immediate business need for churn reduction while offering methodological insights applicable to other telecommunications markets.

Keywords: Random Forest, Customer Churn, Telecommunication, Customer retention

ACRONYMS

1. ANNs – Artificial Neural Networks
2. API - Application Programming Interface
3. AUC - Area Under the Curve
4. CNNs – Convolutional Neural Networks
5. CRISP-DM - Cross-Industry Standard Process for Data Mining
6. CRM- Customer Relationship Management
7. DL – Deep Learning
8. DT – Decision Tree
9. ERT – Extra Random Trees
10. FT – Functional Trees
11. GBM – Gradient Boosting
12. k-NN – k-Nearest Neighbors
13. LightGBM - Light Gradient Boosting Machine
14. LMT – Logistic Model Tree
15. LR – Logistic Regression
16. ML – Machine Learning
17. RF – Random Forest
18. ROC - Receiver Operating Characteristic
19. ROI - Return on Investment
20. SMOTE - Synthetic Minority Over-sampling Technique

Contents

1. CHAPTER 1: INTRODUCTION

1.1 Background

Customer churn defined as customers discontinuing their subscription or services in favor of competing offers represents a critical and persistent challenge within the global telecommunications industry. Churn profoundly affects business profitability, operational sustainability, and long-term market competitiveness. Globally, annual churn rates in telecommunications range between 20% and 40%, contributing to substantial revenue losses for service providers each year and underscoring the urgency for proactive prediction and retention strategies, (*Annual Churn Rate in Global Telecommunications Sector*, 2023).

From an economic perspective, customer retention is significantly more cost-effective than acquisition. Studies indicate that acquiring a new customer can be five to seven times more expensive than retaining an existing one Usman-Hamza et al. (2022). High churn rates can cause inflate acquisition costs, erode market share, reduce revenue stability, and constrain organizational competitiveness. Socially and culturally, stable telecommunication access is essential for enabling critical services, supporting digital inclusion, and facilitating economic participation, particularly in emerging economies such as those in Sub-Saharan Africa *Mobile Economy Report: Sub-Saharan Africa 2022* (2022).

Despite ongoing investment in churn reduction strategies, many telecommunications companies still rely on traditional methods of churn analysis and prediction. These methods often use static and retrospective datasets that inadequately capture dynamic shifts in customer preferences, external socio-economic influences, and seasonal or event-driven fluctuations in usage Chang et al. (2024). Such approaches frequently lead to subjective decision-making, inefficient allocation of retention resources, and limited predictive accuracy.

Recent advances in data science, particularly in Machine Learning (ML), present an opportunity to address these limitations. Predictive analytics, leveraging algorithms such as Logistic Regression, Decision Trees, and Random Forest Classifiers, have demonstrated strong potential for modeling complex customer behaviors and identifying at-risk subscribers before they leave Rahman et al. (2024). These models can integrate large-scale, heterogeneous datasets, uncover non-linear patterns, and generate actionable insights with higher accuracy than conventional approaches.

Telecommunications market is characterized by high competition, rapid digital adoption, and socio-economic reliance on mobile services, these analytical capabilities are particularly relevant. By combining behavioral, demographic, and service usage data with advanced ML techniques, providers can proactively intervene to retain customers, optimize marketing expenditure, and strengthen customer loyalty.

1.2 Research Problem

The telecommunications sector faces intensifying competition, evolving consumer expectations, and increased vulnerability to churn due to competitive offers and service quality perceptions. Current churn prediction and management approaches in many organizations remain reactive, relying on static indicators and manual analyses that are time-consuming, computationally inefficient, and prone to human bias. These methods struggle to capture real-time changes in customer behavior and broader socio-economic shifts, resulting in suboptimal retention strategies and high revenue leakage Mwaura (2021).

1.3 Research Objectives

This study seeks to address the identified gaps through the following objectives:

Main Objective

To predict customer churn in the telecommunication industry.

Specific Objectives

i. To describe the challenges and limitations of existing literature in churn prediction in the telecommunications industry.

ii. To develop a machine learning based churn prediction model using telecommunications data.

iii. To evaluate and compare the performance of machine learning algorithms.

iv. To deploy machine learning models in an interactive dashboard.

1.4 Research Questions

To address the stated objectives, the study is guided by the following research questions:

1. What challenges and limitations exist in current churn prediction approaches within the telecommunications industry?

2. Which demographic, contractual, financial, and usage related factors are the most significant predictors of customer churn?
3. How can machine learning models be developed and optimized to predict churn in the telecommunications sector?
4. How does the performance of different ML algorithms (e.g., Logistic Regression, Random Forest, LightGBM) compare in predicting customer churn?
5. How can a predictive churn model be integrated into a CRM system through an interactive dashboard to support real-time decision-making and interventions?

1.5 Significance and Justification

From a global perspective, effective churn management is critical for sustaining profitability and competitiveness in the telecommunications sector. Retention-focused strategies reduce the high costs associated with customer acquisition as well as stabilize revenue streams, safeguard market share, and enhance long-term customer value.

In emerging markets, the implications of churn extend beyond corporate profitability. Telecommunications services are deeply embedded in daily life, acting as enablers for financial transactions, e-learning, e-health, and small business operations. High churn rates risk excluding segments of the population from these benefits, slowing progress toward digital inclusion and socio-economic development.

This research is justified on three key fronts:

- Economic efficiency: By identifying at-risk customers before they leave, providers can target retention efforts more precisely, thereby reducing costs and maximizing return on marketing investment.
- Operational effectiveness: The integration of the predictive model into CRM systems ensures actionable insights are delivered to decision-makers in real time, enabling timely and effective interventions.
- Scalability and replicability: While tailored to Kenya's socio-economic and competitive environment, the proposed framework offers a methodological blueprint that can be adapted for other markets and industries facing similar churn-related challenges.

This study contributes adapt to both the academic body of knowledge on churn prediction and the practical toolkit available to telecommunications providers.

By leveraging advanced ML techniques and embedding them into operational workflows, the research aims to create a sustainable, data-driven approach to customer retention in highly competitive markets.

2. CHAPTER 2: LITERATURE REVIEW

The telecommunications industry has undergone rapid transformation in recent years due to technological advancement, market liberalization, and shifting consumer expectations. While these trends have enhanced access and service delivery, they have also intensified competition, making customer retention a strategic priority. Customer churn, the phenomenon where subscribers discontinue their relationship with a provider, poses one of the most pressing challenges. Globally, annual churn rates in telecommunication range between 20–40%, resulting in substantial financial losses for service providers and reinforcing the importance of predictive approaches Khan et al. (2024).

2.1 Theoretical Framework

Customer churn has long been viewed as both an economic and relational problem. As noted by Zhang et al. (2022), modern service industries face churn risks driven by service quality and by customer experience and digital interactions. Rogers' Diffusion of Innovation theory remains relevant, explaining why customers adopt, continue, or abandon telecommunication services depending on perceived advantage and compatibility. Likewise, Customer Relationship Management (CRM) frameworks emphasize proactive retention through personalization, loyalty programs, and predictive insights . These theories collectively highlight the multidimensional nature of churn, spanning economic, behavioral, and technological perspectives.

2.2 Empirical Studies

Recent years have witnessed significant advances in churn modeling using machine learning and deep learning. For instance, Usman-Hamza et al. (2022) demonstrated that ensemble decision forests with weighted voting strategies outperformed baseline classifiers in imbalanced telecommunication churn datasets. Similarly, Saha et al. (2023) highlighted the superiority of convolutional neural networks (CNNs) in hybrid datasets from India and the U.S., emphasizing that ROI-sensitive evaluation matters as much as predictive accuracy.

Context-specific models remain essential. In Nepal, Nagarkar (2022) applied XGBoost to 52,332 telecom customers, showing that recharge frequency and inactivity duration are strong predictors of churn, enabling targeted retention campaigns. In Syria, integrated social network analysis (SNA) features, improving AUC from 0.84 to 0.933, demonstrating the value of relational data. More recently, Poudel and Sharma (2024) introduced explainer-aware churn models that combined interpretability methods with predictive algorithms, showing the importance of stakeholder trust in adopting churn systems.

At a broader level,Shahabikargar et al. (2025) reviewed churn prediction across industries and highlighted the growing importance of integrating external and real-time data, as well as the need to balance performance with transparency. Their findings reinforce that churn solutions must be both accurate and interpretable to be actionable in practice.

2.3 Customer Churn in the Telecommunications Industry

Customer churn remains a major challenge for the telecom industry, reducing revenue stability and profitability. Studies confirm that global churn rates can exceed 30%, even in mature markets, highlighting the urgency for robust retention strategies Chang et al. (2024). In Sub-Saharan Africa, where telecoms are a lifeline for financial transactions, e-learning, and small business operations, the stakes are particularly high Mwaura (2021). While predictive modeling has advanced significantly, few studies focus directly on African contexts, revealing a critical gap in geographic representation.

2.4 Industry Relevance and Contribution

This review highlights key insights:

Ensemble and deep learning approaches consistently outperform traditional models.Interpretability and business ROI are increasingly important evaluation criteria, and other markets remain underexplored.

This study propose to contribute by applying comparative machine learning analysis to the Iranian Churn dataset and the Duke Cell2Cell dataset, thereby addressing gaps in cross-market generalization, interpretability, and profit-sensitive evaluation.

The proposed framework provides a scalable and practical approach tailored to the realities of telecommunication churn management in emerging markets.

2.5 Gaps in Existing Literature

- Feature Limitations: Available datasets mainly include structured CRM variables (billing, tenure, usage), but exclude important signals such as network quality, digital engagement, and promotions. This narrows the feature space for building robust churn models Mwaura (2021).

- Evaluation Metrics Bias: Existing research often prioritizes accuracy and AUC, while overlooking profit-sensitive or ROI-based metrics Shahabikargar et al. (2025). This creates a gap between academic results and actionable business insights.
- Interpretability Challenges: High-performing ensemble and deep learning models are frequently treated as "black boxes." Limited interpretability makes it difficult for managers to trust model outputs Chang et al. (2024) and translate them into real retention strategies.

2.6 Conceptual Framework

The conceptual framework for this study comprises the following components:

- User Behavior Analysis: Examination of customer demographics, usage patterns, contractual terms, and payment behaviors to identify churn predictors.
- Machine Learning Models: Implementation of algorithms including Logistic Regression, Random Forest, Decision Trees, and Gradient Boosting to predict churn, with comparative performance analysis.
- User Retention Strategies: Translation of model outputs into actionable retention initiatives tailored to the Kenyan telecommunications market.
- Performance Evaluation: Use of metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to assess predictive performance.
- Contextual Considerations: Incorporation of Kenya's socio-economic and operational realities to ensure recommendations are practical and market-relevant.

3. CHAPTER 3: METHODOLOGY

This study adopts the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology to ensure a structured, repeatable, and scalable approach to customer churn prediction in the Kenyan telecommunications sector.

3.0.1 Proposed Workflow

The proposed workflow for this study is illustrated in Figure 3.1. It outlines the systematic process of building and evaluating machine learning models for customer churn prediction.

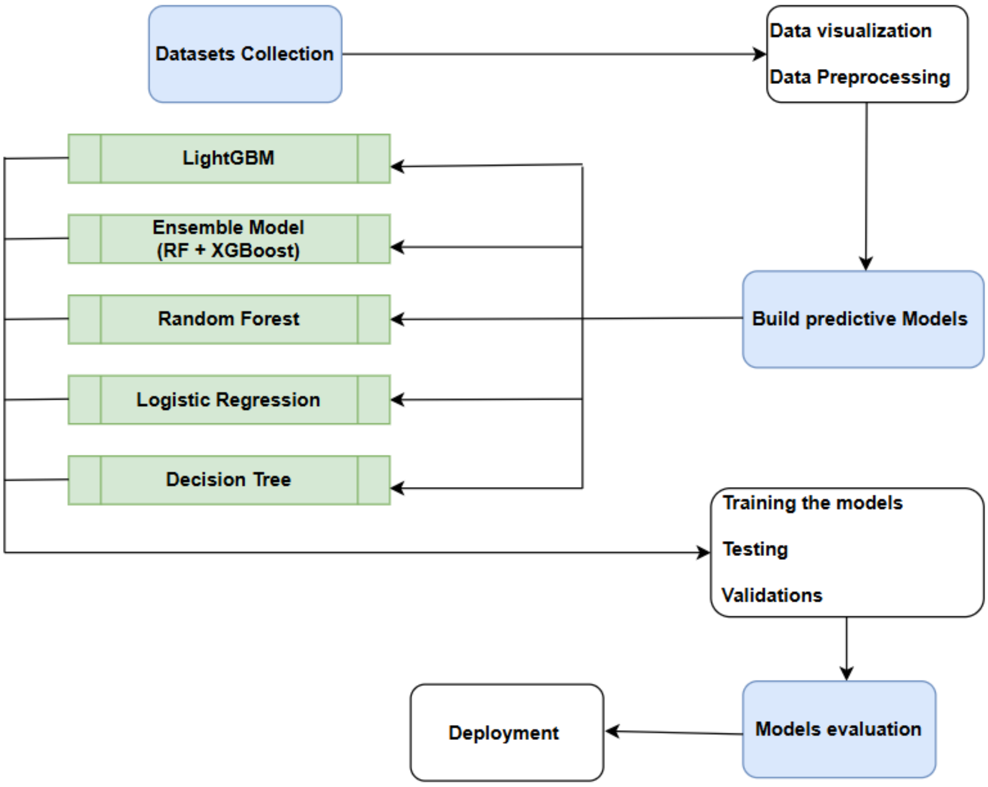

Figure 3.1: Proposed Workflow

3.1 Business Understanding

The telecommunications sector operates in an increasingly competitive environment characterized by market saturation, declining profit margins, and evolving consumer expectations. In such a landscape, customer retention has become a critical determinant of long-term profitability and business sustainability. The cost of acquiring new customers often exceeds that of retaining existing ones, making churn reduction a strategic priority for telecom operators. However, despite heavy investments in marketing and customer engagement, many providers continue to face substantial churn rates that erode revenue and weaken market share.

In the Kenyan telecommunications market, competition among major service providers such as Safaricom, Airtel, and Telkom Kenya has intensified, driven by aggressive pricing strategies, rapid technological adoption, and diverse value-added services. Customers frequently switch between operators in search of better data plans, network quality, or customer support, leading to unpredictable churn behavior. Traditional approaches to churn management within these companies are largely reactive, relying on historical or static data that fail to capture real-time behavioral dynamics and socio-economic influences. This has resulted in inefficient resource allocation and delayed interventions that often occur after the customer has already defected.

3.2 Data Understanding

This study will employ secondary datasets widely used in churn prediction research: the Iranian Churn dataset (UCI Repository) and the Duke Cell2Cell dataset (Teradata Center, Duke University). The Iranian dataset captures attributes such as customer demographics, subscription type, and service usage patterns, while the Duke Cell2Cell dataset includes contractual information, billing records, and behavioral variables. Together, these datasets provide a comprehensive representation of churn dynamics across different telecommunication markets.

3.3 Data Collection

This study will utilize secondary data obtained from two well-established telecommunications churn datasets: the Iranian Churn dataset, hosted by the UCI Machine Learning Repository, and the Duke Cell2Cell dataset, available through the Teradata Center for Customer Relationship Management at Duke University. The Iranian dataset provides customer-level attributes such as demographics, subscription types, and service usage, while the Duke Cell2Cell dataset offers extensive behavioral, contractual, and financial information on telecommunication customers.

The use of these datasets ensures access to large-scale, high-quality, and pre-validated records that capture diverse aspects of customer engagement and churn behavior. Importantly, both datasets allow for comparative analysis across different market contexts, strengthening the generalizability of findings. All data are fully anonymized and publicly available, ensuring compliance with ethical standards and data governance requirements. By leveraging these datasets, the study gains a robust foundation for developing, testing, and validating machine learning models for churn prediction, while addressing existing gaps in literature related to transferability and cross-market benchmarking.

3.4 Exploration Data Analysis

Through data exploration, this study will derive valuable insights into the distribution, variability, and structure of the collected dataset. This phase will involve performing descriptive statistical analyses, generating visualizations, and computing summary statistics to identify patterns, trends, and potential relationships within the data. Exploratory Data Analysis (EDA) techniques, including histograms, scatter plots, box plots, and correlation matrices, will be used to assess the relationships between customer demographic factors, contractual attributes, financial metrics, and service usage patterns. These insights will inform the selection of appropriate machine learning models and preprocessing strategies.

3.5 Data Cleaning

Data cleaning will constitute a critical component of the data pre-processing phase, undertaken to enhance the quality, accuracy, and reliability of the dataset prior to analysis. Given that the data will be obtained through structured questionnaires, potential sources of error may include typographical mistakes, incomplete responses, inaccurately recorded values, and inconsistencies in categorical labels or formatting.

A systematic cleaning process will be applied to address these issues. Specifically, missing values will be imputed using appropriate statistical or rule-based methods to preserve the representativeness and distributional characteristics of the data. Erroneous entries and illogical values will be identified and corrected through consistency checks and cross-validation against predefined logical rules. Furthermore, categorical and numerical variables will be standardized to ensure uniformity across all responses, thereby facilitating compatibility with subsequent analytical procedures.

This rigorous data cleaning protocol is intended to minimize noise, reduce the risk of bias, and ensure that the final dataset is robust, coherent, and fit for reliable statistical analysis and predictive modeling.

3.5.1 Treating Missing Data

Addressing missing data is critical to ensuring the completeness and quality of the dataset. Missing values will be handled based on their nature and extent. Where feasible, imputation techniques such as mean imputation, regression-based imputation, or domain-informed replacement will be applied. Where the proportion of missing data is excessive or the missingness is deemed non-random, the affected entries may be excluded to prevent bias. The choice of imputation method will be guided by the variable type, data distribution, and potential impact on predictive performance.

3.5.2 Treating Outliers

Outliers have the potential to distort statistical summaries and reduce the accuracy of predictive models. Outlier detection will be conducted using statistical thresholds (e.g., z-scores, interquartile ranges) and visual diagnostics (e.g., box plots, scatter plots). Detected outliers will be evaluated in consultation with subject matter experts to determine whether they represent valid extreme cases or erroneous entries. Appropriate treatments, such as capping, transformation, or removal, will be applied as necessary.

3.5.3 Data Type Conversion

Data type conversion will ensure that variables are consistently represented for analysis. Categorical variables will be converted into standardized label formats, and numerical variables will be cast to appropriate numeric data types. This process ensures compatibility with machine learning algorithms and facilitates reproducibility.

3.5.4 Data Transformation

Feature Engineering

The dataset will be refined to improve model predictive capacity through feature engineering. This process will involve generating new variables or modifying existing ones to capture latent patterns in customer churn behavior. Examples may include the computation of customer tenure categories, normalized spending ratios, and derived usage intensity scores.

Feature Scaling

Feature scaling will standardize the range of numerical variables to ensure equitable influence on model training. Standardization using z-score normalization and normalization using Min-Max scaling will be considered. These methods prevent variables with large numerical ranges (e.g., monthly charges) from dominating the learning process.

Feature Selection

Relevant features will be selected to reduce dimensionality, mitigate overfitting, and improve interpretability. Statistical tests, correlation analysis, and model-based feature importance rankings will guide the selection process. Irrelevant or redundant features will be excluded to enhance computational efficiency.

Data Encoding

Categorical attributes will be transformed into numerical representations suitable for machine learning models. One-hot encoding, label encoding, or binary encoding will be applied depending on the nature of the variable and the target algorithm's requirements.

3.6 Modeling

Customer churn prediction constitutes a binary classification task, requiring the accurate categorization of customers into *churn* and *non-churn* groups based on their demographic, contractual, financial, and behavioral attributes. To address this challenge, the study adopts a comparative modeling approach that evaluates several machine learning algorithms, each chosen for its unique strengths and suitability for the churn prediction problem in the telecommunications sector.

3.6.1 Logistic Regression (LR)

Logistic Regression serves as the foundational benchmark model due to its simplicity, computational efficiency, and interpretability. It estimates the probability that a customer i will churn ($y_i = 1$) given a vector of predictors $\mathbf{x}_i = [x_{i1}, x_{i2}, \ldots, x_{ip}]$ through the logistic (sigmoid) function:

$$P(y_i = 1 \mid \mathbf{x}_i) = \sigma(\mathbf{w}^\top \mathbf{x}_i + b) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x}_i + b)}} \tag{3.1}$$

where $\mathbf{w}$ represents the vector of feature coefficients and b is the bias term. Model parameters are estimated by maximizing the log-likelihood:

$$\mathcal{L}(\mathbf{w}, b) = \sum_{i=1}^{N} [y_i \log P(y_i) + (1 - y_i) \log(1 - P(y_i))] \tag{3.2}$$

Logistic Regression provides clear insights into how each feature influences churn probability, which aids communication with non-technical stakeholders.

3.6.2 Decision Tree (DT)

Decision Trees partition the input space into hierarchical, non-overlapping regions by recursively splitting the data based on feature values that minimize an impurity measure. The most common splitting criterion for classification is the *Gini Index*:

$$\text{Gini}(t) = 1 - \sum_{k=1}^{K} p_k^2 \tag{3.3}$$

where p_k denotes the proportion of samples belonging to class k at node t. At each node, the feature and threshold that minimize the weighted impurity across child nodes are chosen:

$$\Delta\text{Gini} = \text{Gini}(t) - \sum_{j \in \text{children}} \frac{N_j}{N_t} \text{Gini}(j) \tag{3.4}$$

Decision Trees are favored for their interpretability and ability to visualize decision rules in a transparent manner.

3.6.3 Random Forest (RF)

Random Forests extend Decision Trees by constructing an ensemble of B trees, each trained on a bootstrap sample of the data and a random subset of features. The final prediction $\hat{y}$ for a given observation $\mathbf{x}$ is obtained through majority voting:

$$\hat{y} = \text{mode}\{h_b(\mathbf{x})\}_{b=1}^{B} \tag{3.5}$$

where $h_b(\mathbf{x})$ is the class prediction from the b^{th} tree. Random Forests reduce variance and mitigate overfitting while maintaining high predictive accuracy and feature interpretability.

3.6.4 Light Gradient Boosting Machine (LightGBM)

LightGBM is a gradient boosting framework that builds additive models in a forward stage-wise manner. At each iteration m, a weak learner $h_m(\mathbf{x})$ (typically a shallow tree) is fitted to the negative gradient of the loss function $\mathcal{L}$ with respect to the current model prediction $F_{m-1}(\mathbf{x})$:

$$r_{im} = -\left[\frac{\partial \mathcal{L}(y_i, F_{m-1}(\mathbf{x}_i))}{\partial F_{m-1}(\mathbf{x}_i)}\right] \tag{3.6}$$

The model is then updated as:

$$F_m(\mathbf{x}) = F_{m-1}(\mathbf{x}) + \eta\, h_m(\mathbf{x}) \tag{3.7}$$

where η is the learning rate. LightGBM optimizes this process through histogram-based algorithms and leaf-wise tree growth, enabling scalability for large datasets with high-dimensional features.

3.6.5 Ensemble Model

To enhance robustness and stability, an ensemble model combines predictions from multiple base learners (e.g., Logistic Regression, Random Forest, and LightGBM). The combined prediction $\hat{y}_{\text{ensemble}}$ can be expressed as a weighted average of individual model probabilities:

$$\hat{y}_{\text{ensemble}} = \sum_{m=1}^{M} \alpha_m \, \hat{y}_m \tag{3.8}$$

where α_m denotes the weight assigned to model m and $\sum_{m=1}^{M} \alpha_m = 1$. This aggregation reduces individual model bias and variance, leading to improved generalization performance.

3.6.6 Handling Class Imbalance with SMOTE

Customer churn datasets often exhibit significant class imbalance, with churners constituting a small fraction of the total population. To mitigate this issue, the Synthetic Minority Over-sampling Technique (SMOTE) will be applied to generate synthetic minority samples. For each minority instance x_i, a synthetic sample x_{new} is created as:

$$x_{\text{new}} = x_i + \delta \times (x_{nn} - x_i), \quad \delta \sim U(0,1) \tag{3.9}$$

where x_{nn} is one of the k nearest neighbors of x_i. This approach balances the training data distribution, enabling the Logistic Regression model trained on SMOTE-augmented data to achieve better sensitivity in detecting churners.

3.7 Model Evaluation

The performance of all predictive models will be assessed using an 80/20 train–test split, implemented with stratified sampling to preserve the proportional representation of churn and non-churn classes. Model effectiveness will be evaluated using standard classification metrics, namely *Precision*, *Recall*, and the *F1-Score*, defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad \text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.10}$$

where TP, FP, and FN represent the number of true positives, false positives, and false negatives, respectively. Precision quantifies the proportion of correctly predicted churners among all predicted churners, while recall measures the proportion of actual churners that were successfully identified. The F1-Score serves as the harmonic mean of precision and recall, balancing the trade-off between the two metrics.

In addition, the Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) will be computed to evaluate the model's overall discriminatory power. A higher AUC value indicates better separation between churners and non-churners. Given the substantial business cost associated with false negatives (i.e., failing to identify actual churners), recall will be prioritized during model selection and ranking to ensure effective customer retention strategies.

3.8 Model Deployment

The selected model will be deployed via an Application Programming Interface (API) to enable integration with existing customer management systems. Deployment through an API will provide modularity, maintainability, and scalability, allowing the predictive engine to serve multiple client applications concurrently.

Deployment Framework: FastAPI will be used as the deployment framework due to its speed, built-in data validation, automatic documentation, and asynchronous capabilities. The deployed model will return churn probability scores and key predictive factors for each customer record. These outputs will support targeted retention campaigns and ongoing monitoring of churn risk.

4. CHAPTER 4: EXPECTED OUTCOMES

The proposed study aims to develop a data-driven, machine learning framework for predicting customer churn in the telecommunications industry. Using secondary datasets from the Iranian Churn (UCI Repository) and Duke Cell2Cell (Teradata CRM Center), the research will produce actionable insights applicable to emerging markets such as Kenya. The outcomes will address key research gaps, including cross-market generalization, interpretability, and profit-sensitive evaluation.

To address the first objective, the study will identify and analyze existing gaps in churn prediction literature, particularly the lack of diverse features, limited interpretability, and weak model transferability. For the second objective, predictive models will be developed to capture critical behavioral, financial, and contractual churn determinants, integrating features such as billing trends, service usage, and demographics.

The third objective involves evaluating and comparing algorithms including Logistic Regression, Decision Tree, Random Forest, and LightGBM using metrics such as recall, precision, F1-score, and ROC-AUC to determine the most effective and interpretable model. In line with the fourth objective, an interactive dashboard will be deployed using explainable AI techniques (e.g., SHAP values) to visualize churn drivers and support real-time, data-informed decision-making.

The study is expected to deliver a scalable and interpretable churn prediction system that enhances proactive retention strategies, minimizes revenue loss, and improves operational efficiency. Beyond the Kenyan context, the framework will serve as a replicable model for other African telecommunications markets, promoting data-driven decision-making and customer-centric business sustainability.

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